9.7 Power Series

"polynomial of infinite degree"

From infinite series of numbers, we can study "infinite polynomials" - **power series**.

Many functions (e.g. e^x, $\ln x$, $\sin x$, etc) have useful representations as a power series.

Definition: A **power series about** $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \ldots + c_n x^n + \ldots.$$

A **power series about** $x = a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \ldots + c_n (x-a)^n + \ldots$$

in which the **center** a and the coefficients c_o, c_1, c_2, ..., c_n, ... are constants.

Aim: For a function $f(x)$, we want to write

$f(x)$ "$=$" $\sum_{n=0}^{\infty} c_n (x-a)^n$

Example: The geometric power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots + x^n + \ldots$$

converges to $1/(1 - x)$ for $|x| < 1$. This is usually express by writing

$$\frac{1}{1-x} = 1 + x + x^2 + \ldots + x^n + \ldots, \quad -1 < x < 1.$$

Note: The partial sums of the series are the polynomials $P_n(x)$ that approximate the function on the left. The approximations do not apply when $x \geq 1$.

$P_0(x) = 1$

$P_1(x) = 1 + x$

$P_2(x) = 1 + x + x^2$

$\vdots$

$P_n(x) = 1 + x + x^2 + \ldots + x^n$

for $|x| < 1$, we have

$$\lim_{n \to \infty} P_n(x) = \frac{1}{1-x}.$$

Example: Consider the power series

$$1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \ldots + \left(-\frac{1}{2}\right)^n (x-2)^n + \ldots$$

This is a geometric series with the first term = 1 & ratio $r = -\frac{1}{2}(x-2)$

The series converges for $|r| < 1$

$$\Rightarrow \left| -\frac{1}{2}(x-2) \right| < 1$$

$$\Rightarrow 0 < x < 4 .$$

So the series $\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (x-2)^n = \frac{1}{1+\frac{1}{2}(x-2)} = \frac{2}{x}$ for $0 < x < 4$

We write $\frac{2}{x} = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (x-2)^n$, $0 < x < 4$.

We can test the convergence of a power series using the Ratio Test.

Example: For what value of x do the following power series converge?

(a). $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ $\quad -1 < x \leq 1$

To check where $\sum \boxed{(-1)^{n-1} \frac{x^n}{n}}$ ($= \sum u_n$) converges, we look at $\sum |u_n|$ instead.

$|u_n| = \frac{x^n}{n}$ Ratio Test: $\frac{|u_{n+1}|}{|u_n|} = \frac{|x^{n+1}/n+1|}{|x^n/n|} = |x| \cdot \frac{n}{n+1}$

$$\lim_{n\to\infty} \frac{|u_{n+1}|}{|u_n|} = |x|$$

Now, by the Ratio Test, if $|x| < 1$, then the series $\sum |u_n|$ converges
$\Rightarrow \sum u_n$ also converges.

For $|x| > 1$, the nth term test $\Rightarrow$ the series diverges.

For $x = 1$, this is an alternating harmonic series $\Rightarrow$ so converges.
For $x = -1$, we have $-\sum \frac{1}{n}$, clearly a divergent series.

(b). $\sum_{n=1}^{\infty} \underbrace{(-1)^{n-1} \frac{x^{2n-1}}{2n-1}}_{u_n} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$

$$\frac{|u_{n+1}|}{|u_n|} = \frac{|x^{2n+1}/(2n+1)|}{|x^{2n-1}/(2n-1)|} = \frac{2n-1}{2n+1} x^2$$

$$\lim_{n\to\infty} \frac{|u_{n+1}|}{|u_n|} = x^2$$

By the Ratio Test, the series converges absolutely (hence converges) for $-1 < x < 1$.

For $|x| > 1$, the nth term $\not\to 0$ so diverges.

For $x = \pm 1$, it is an alternating series, and can check that it converges. (AST)

Convergence: $-1 \le x \le 1$

(c). $\sum_{n=0}^{\infty} \boxed{\frac{x^n}{n!}} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ $\quad : \sum_{n=0}^{\infty} x^n/n! \; = e^x$

$\frac{|u_{n+1}|}{|u_n|} = \frac{|x^{n+1}/(n+1)!|}{|x^n/n!|} = \frac{|x|}{n+1} \Rightarrow \lim_{n\to\infty} \frac{|u_{n+1}|}{|u_n|} = 0$

So the Ratio Test $\Rightarrow$ the series converges at all $x \in \mathbb{R}$.

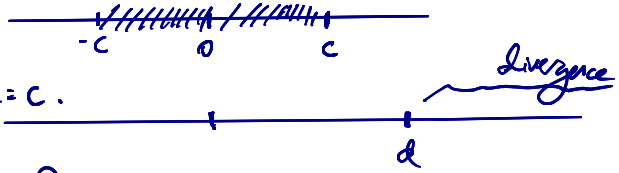
(d). $\sum_{n=1}^{\infty} \boxed{n!\, x^n} = 1 + x + 2!\, x^2 + 3!\, x^3 + \ldots$

$\frac{|u_{n+1}|}{|u_n|} = \frac{|(n+1)!\, x^{n+1}|}{|n!\, x^n|} = (n+1)|x| \longrightarrow \infty$ if $x \neq 0$.

so the series diverges everywhere except for $x = 0$.

Theorem (The Convergence Theorem for Power Series):

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \ldots$ converges at $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. If the series diverges at $x = d$, then it diverges for all x with $|x| > |d|$.



Proof: Suppose $\sum a_n x^n$ converges at $x = c$.

Then the nth term test $\Rightarrow \lim_{n\to\infty} a_n c^n = 0$.

There exists N s.t. for all $n \geq N$,

$$|a_n c^n| < 1 \Rightarrow |a_n| < \frac{1}{|c|^n}.$$

Now let x be s.t. $|x| < |c|$ & consider the power series $\sum a_n x^n$.

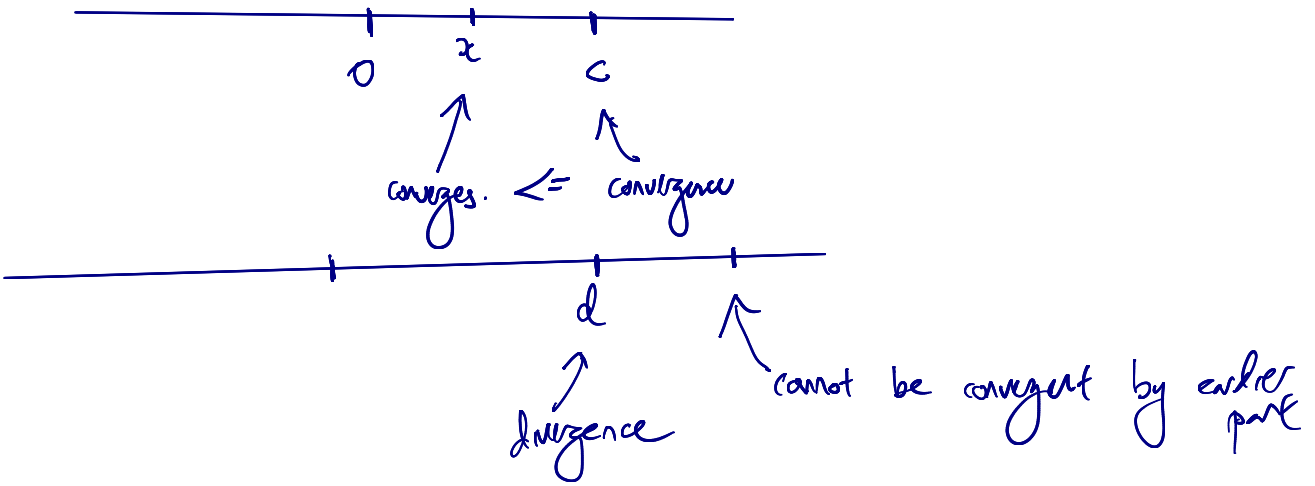
Observe that for all $n \geq N$,

$$|a_n x^n| = |a_n||x|^n < \frac{1}{|c|^n} \cdot |x|^n = \left(\frac{|x|}{|c|}\right)^n$$

$\Rightarrow \sum_{n=N}^{\infty} \left(\frac{|x|}{|c|}\right)^n$ converges (geometric series with ratio < 1)

DCT $\Rightarrow \sum_{n=N}^{\infty} |a_n x^n|$ converges.

$\Rightarrow \sum a_n x^n$ converges.



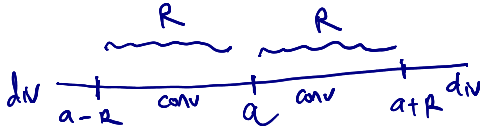
Suppose $\sum a_n d^n$ diverges, let $x > d$. $(d > 0)$

If $\sum a_n x^n$ converges, then the first part of the theorem says that $\sum a_n d^n$ would have to be convergent, which contradicts to the assumption that $\sum a_n d^n$ diverges.

Therefore, $\sum a_n x^n$ diverges.

The Radius of Convergence of a Power Series

Corollary: The convergence of the series $\sum c_n(x-a)^n$ is described by one of the following three cases:



1. There is a positive number R such that the series diverges for x with $|x-a| > R$ but converges absolutely for x with $|x-a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.

2. The series converges absolutely for every x $(R = \infty)$.

3. The series converges at $x = a$ and diverges elsewhere $(R = 0)$.

R is called the **radius of convergence** of the power series, and the interval of radius centered at $x = a$ is called the **interval of convergence**.

radius = R

interval of convergence

a-R, a+R

- [a-R, a+R]
- (a-R, a+R)
- [a-R, a+R)
- (a-R, a+R]

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

converges at $x=a$ only.
$(R=0)$

Use Ratio Test (or Root Test) for absolute convergence to obtain.
$|x-a|<R$

converges absolutely for all $x \in \mathbb{R}$
$(R=\infty)$

Then substitute $x = a+R$ & $x = a-R$

to get $\sum c_n R^n$ & $\sum (-1)^n c_n R^n$

then test convergence using convergence tests.

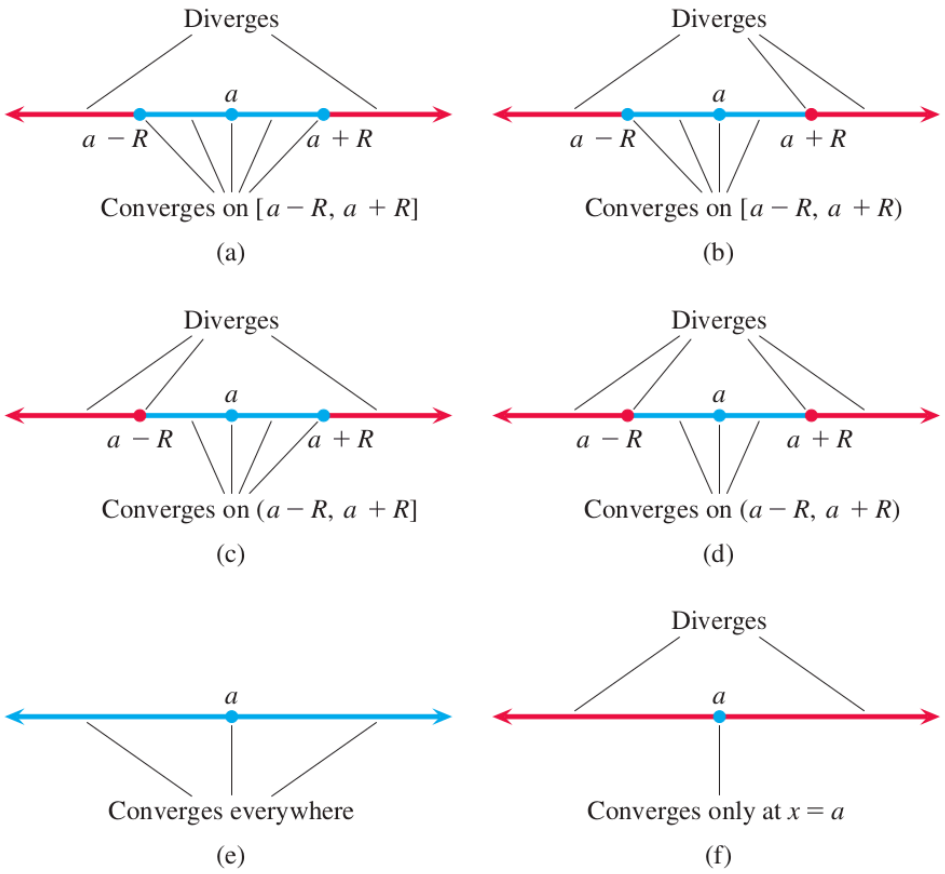


FIGURE 9.21 The six possibilities for an interval of convergence.

How to Test a Power Series for Convergence

1. *Use the Ratio Test (or Root Test) to find the interval where the series converges absolutely.* Usually, this is an open interval

$$|x - a| < R \qquad \text{or} \qquad a - R < x < a + R.$$

2. *If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint.* Use a Comparison Test, the Integral Test, or the Alternating Test.

3. *If the interval of absolute convergence is* $a - R < x < a + R$*, the series diverges for* $|x - a| > R$ (it does not even converge conditionally) because the nth term does not approach zero for those values of x.

Examples:

1. $\sum_{n=0}^{\infty} \boxed{\frac{(x-2)^n}{10^n}}$ u_n

2. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x+2)^n}{n2^n}$

(1) Check absolute convergence : $\sum |u_n|$

Ratio Test : $\frac{|u_{n+1}|}{|u_n|} = \frac{|(x-2)|}{10}$

Converges when $\frac{|x-2|}{10} < 1$

$\Rightarrow -8 < x < 12$

$\Rightarrow$ Radius of Convergence $R = 10$.

- $x = -8 \Rightarrow \sum u_n = \sum (-1)^n$ so diverges.
- $x = 12 \Rightarrow \sum u_n = \sum 1^n$ so diverges.

Interval of convergence $= (-8, 12)$.

(2) $\sum_{n=1}^{\infty} \boxed{\frac{(-1)^{n+1}(x+2)^n}{n\,2^n}}\ u_n$

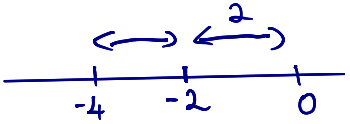
$\sum |u_n|$: $\dfrac{|u_{n+1}|}{|u_n|} = \dfrac{|(x+2)^{n+1}/(n+1)2^{n+1}|}{|(x+2)^n|/n2^n}$

$= \dfrac{|x+2|}{2} \cdot \dfrac{n}{n+1}$

$\longrightarrow \dfrac{|x+2|}{2}$ as $n \to \infty$.

- When $|x+2| > 2$, the power series diverges by the nth term test.

Ratio Test $\Rightarrow$ Converges when $\dfrac{|x+2|}{2} < 1 \Rightarrow -4 < x < 0$



Radius of convergence $R = 2$.

- $x = -4$: $\sum u_n = \sum \dfrac{-1}{n} = (-1)\sum \dfrac{1}{n}$ diverges.
- $x = 0$: $\sum u_n = \sum_{n=1}^{\infty} (-1)^{n+1} \dfrac{1}{n}$ converges.

$\Rightarrow$ Interval of convergence : $(-4, 0]$.

Operations on Power Series

Theorem (The Series Multiplication Theorem for Power Series): If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$, and

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \ldots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k}$$

$c_n = 1 + 1 + 1 + \ldots + 1 = n+1$

then $\sum_{n=0}^{\infty} c_n x^n$ converges absolutely to $A(x)B(x)$ for $|x| < R$:

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n.$$

Eg. If $A(x) = \sum_{n=0}^{\infty} x^n$ & $B(x) = \sum_{n=0}^{\infty} x^n$, then $A(x)$ & $B(x)$ converges absolutely for $|x| < 1$.

$A(x)B(x) = \left(\sum_{n=0}^{\infty} x^n\right)\left(\sum_{n=0}^{\infty} x^n\right) = (1 + x + x^2 + \ldots)(1 + x + x^2 + \ldots)$

$= 1 + 2x + 3x^2 + \ldots + (n+1)x^n + \ldots$

$= \sum_{n=0}^{\infty} (n+1) x^n$

converges absolutely for $|x| < 1$.

We can also substitute a function $f(x)$ for x in a convergent power series.

Theorem: If $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for $|x| < R$, then $\sum_{n=0}^{\infty} a_n (f(x))^n$ converges absolutely for any continuous function f on $|f(x)| < R$.

Example: $1/(1 - 4x^2) = \sum_{n=0}^{\infty} (4x^2)^n$

The power series $\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n$ converges absolutely when $|x| < 1$.

Setting $f(x) = 4x^2$, we have

$$\frac{1}{1-4x^2} = 1 + 4x^2 + (4x^2)^2 + \cdots = \sum_{n=0}^{\infty} (4x^2)^n .$$

converges for $|4x^2| < 1 \Rightarrow |x| < \frac{1}{2}$.

Theorem (The Term-by-Term Differentiation Theorem): If $\sum c_n(x-a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n \text{ on the interval } a - R < x < a + R.$$

The function f has the derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1},$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)c_n(x-a)^{n-2},$$

and so on, Each of these derived series converges at every point of the interval $a - R < x < a + R$.

Example: Find series for $f'(x)$ and $f''(x)$ if

$f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots, \quad -1 < x < 1.$

$f'(x) = \frac{1}{(1-x)^2} \overset{\text{Thm}}{=} 1 + 2x + 3x^2 + 4x^3 + \ldots = \sum_{n=0}^{\infty} (n+1)x^n$

converges for $-1 < x < 1$

$f''(x) = \frac{2}{(1-x)^3} = 2 + 3 \cdot 2x + 4 \cdot 3x^2 + 5 \cdot 4x^3 + \ldots = \sum_{n=0}^{\infty} (n+2)(n+1)x^n$

↑ Thm ⇒ converges for $-1 < x < 1$.

A power series can also be integrated term by term throughout its interval of convergence.

Theorem (The Term-by-Term Integration Theorem): Suppose that $f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n$ converges for $a - R < x < a + R \quad (R > 0)$. Then

$$\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

converges for $a - R < x < a + R$ and

$$\int f(x)\, dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C$$

for $a - R < x < a + R$.

Tutorial : $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{1}{n}\right)$ vs $\sum \frac{1}{n}$

Example: Identify the function

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots, \quad -1 \le x \le 1.$$

Observe that $f(0) = 0$.

Note that $f'(x) = 1 - x^2 + x^4 - x^6 + x^8 - \ldots \quad -1 < x < 1$

↑ geometric series with ratio $-x^2$

$$= \frac{1}{1-(-x^2)} = \frac{1}{1+x^2}$$

Thm $\Rightarrow$ $f(x) = \int f'(x)\,dx = \int \frac{1}{1+x^2}\,dx = \tan^{-1}(x) + C$

$f(0) = \tan^{-1}(0) + C \Rightarrow C = 0$

$\tan^{-1}\left(\frac{1}{n}\right) = \frac{1}{n} - \frac{(1/n)^3}{3} + \frac{(1/n)^5}{5} - \ldots$

$\therefore$ $f(x) = \tan^{-1}(x)$ $\left[= x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots \right]$

Example: Consider the series

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \ldots \quad , \quad -1 < t < 1$$

$$= \sum_{n=0}^{\infty} (-1)^n t^n .$$

We can write.

$$\ln(1+x) = \int_0^x \frac{1}{1+t}\, dt .$$

$$= \int_0^x \sum_{n=0}^{\infty} (-1)^n t^n \, dt .$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{(n+1)} \quad , \quad -1 < x < 1 .$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \ldots .$$